# Object-Centric Mask Prompts for a Frozen Robot Policy: Pixel-Fidelity Verification and Preregistered Null Results for Fine-Region and Mask Cues

Anonymous CVPR submission

## Abstract

*Can a frozen robot policy benefit from a more precise description of what to grasp and where to place it? We study this question for one frozen vision-language-action policy on one RoboCasa task under a paired protocol in which only the image/text presentation differs between arms. First, a preregistered comparison of the existing point cue against a fine-region ("neighborhood") cue on 12 unused seeds found no evidence of benefit: under the registered accounting rule both arms succeeded on the same 6 of 10 admissible pairs (0 discordant pairs; difference 0.0 percentage points, conservative 95% interval [−35.5, 35.5], exact paired $p = 1$), and a disclosed deviation-inclusive analysis gives 7/11 for points and 6/11 for neighborhoods (−9.1 points, [−45.7, 32.7], $p = 1$). Second, we present MaskCue, a stage-conditioned interface that draws two-pixel exterior contours of the exact visible object mask and of a subject-sized destination proxy in a dedicated single-sample segmentation context, so that object interiors are never painted, and verify its pixel fidelity on an NVIDIA H800 and again on the NVIDIA H100 that ran the study: 11 of 12 predeclared checks passed (0 mismatched pixels against an independent reference render, 0 interior pixels changed, destination-proxy cross-check IoU $\geq 0.9925$); the failing check (3 pixels outside a 1-pixel band in 1 of 117 masks, the same mask on both GPUs) and a measured 1-pixel boundary bleed from the policy's own bilinear resize are disclosed. Third, the preregistered 12-pair mask-versus-point comparison, whose first attempt was lost with its H800 instance before any outcome existed, was executed on a shared H100 cluster with the same frozen sources, seeds, pairs, and analyzer. All 12 pairs were admissible: the point cue succeeded in 8/12 and the mask cue in 5/12 (difference −25.0 points, conservative 95% interval [−74.1, 39.7], exact paired $p = 0.45$). Neither study supports a success benefit from a more precise spatial cue or establishes equivalence; every deviation between registration and execution is recorded. Masks use simulator geometry: this is an oracle-assisted presentation study, not RGB-only perception.*

## 1. Introduction

An instruction such as "place the mug under the dispenser" leaves a learned robot policy to infer both the manipulated instance and its destination. Visual prompts can make those references explicit, but greater geometric detail need not produce better actions: a contour can hide a thin handle, and a cue that is exterior in the raw image can blend into the object after the policy's own resizing. We therefore separate *pixel fidelity*, whether the displayed cue is consistent with the declared current geometry and survives the policy's preprocessing, from *task utility*, whether the cue changes task success under a paired protocol; a high success rate cannot excuse a prompt that uses undeclared simulator information. This paper reports what we could establish for one frozen vision-language-action policy, the RoboCasa365 checkpoint of Xiaomi-Robotics-1 [9], wrapped by a frozen Qwen3-VL-4B-Instruct stage planner and verifier [1], on the RoboCasa `CoffeeSetupMug` task [5, 7]; only the image/text presentation differs between arms.

**A completed null result for fine-region cues.** Before building the mask interface, we ran a preregistered paired comparison of the existing point cue against a fine-region "neighborhood" presentation on 12 unused seeds. Under the registered accounting rule both cues succeeded on the same six of ten admissible pairs; a disclosed supplement including one re-executed zero-step case favors the point control by one success among eleven pairs. Neither analysis supports a benefit or establishes equivalence; the result is never pooled with the mask study.

**A verified mask interface.** MaskCue is an inference-time presentation interface for a frozen, chunked policy. During grasp it draws a contour just outside the selected object's visible mask; during placement it outlines the held subject and a subject-sized destination proxy (a "ghost box") at the bound placement anchor. It changes only images and a fixed explanatory sentence, injecting no motion primitive and overriding no command; Sec. 3 describes its single-sample rendering context, protected interiors, painter cross-

check, and crop/resize audit. On both evaluation GPUs, 11 of 12 predeclared pixel-level checks passed; the one failure and the boundary bleed after bilinear resizing are reported, not repaired after the fact.

**A completed null result for mask cues.** We froze 12 further unused seeds, a 24-case manifest, ten pilot gates, and the same analysis with masks in place of neighborhoods. After the first pilot was lost with its H800 instance, the study ran on a shared H100 cluster under the same frozen protocol (Sec. 4). Every pair was admissible: the point cue succeeded in 8 of 12 pairs and the mask cue in 5 of 12 ($-25.0$ points, conservative 95% interval $[-74.1, 39.7]$, exact paired $p = 0.45$), supporting no benefit from exact contours and no equivalence.

**Contributions.** (1) A preregistered negative result for fine-region cues with complete accounting of unknown cases and a disclosed deviation (Sec. 5.1). (2) A stage-conditioned object/destination mask interface verified on two GPUs with predeclared gates, including a failed check and a measured limit of raw-frame exteriority under the policy's preprocessing (Secs. 3 and 5.2). (3) A preregistered mask-versus-point comparison with no unknown pair, a negative point estimate, and an explicit ledger of every deviation between registration and execution (Secs. 4 and 5.3). Whether any presentation improves control for this policy, or generalizes beyond one task, remains open.

## 2. Related Work

**Visual prompting.** Set-of-Mark prompting exposes regions to multimodal models through explicit marks [10]; MOKA combines task decomposition with marked keypoints and grids for manipulation [2]; PIVOT iteratively renders spatial proposals for a vision-language model to select [6]. Our claim is narrower: exterior object and destination contours at the input of an unchanged policy, under declared information access and pixel-fidelity checks; drawing a segmentation contour is not itself a new principle.

**Geometric and hierarchical control.** ReKep optimizes trajectories under spatiotemporal constraints over 3D keypoints [3]; Hi Robot connects high-level reasoning to low-level vision-language-action control [8]. Our proxy is projected into images, not converted into an optimized trajectory or analytic controller, and the planner and verifier are shared by both arms, so the comparison does not isolate hierarchical planning.

**Frozen-policy harnesses.** RoboHarness studies memory and in-context adaptation around frozen policies [4]; Harness VLA combines policy primitives, analytic skills, and memory [11]. We restrict the intervention to a declared image/text presentation evaluated against the same wrapper with point cues, so results from other embodiments, tasks, or information budgets are not directly comparable.

## 3. Stage-Conditioned Mask Interface

### 3.1. Shared execution system

Let $I_t$ be the current multi-camera RGB observation, $q_t$ native proprioception, and $u_k$ the instruction of active stage $k$, which the shared planner binds to a persistent entity handle (Fig. 1). A presentation function $F_a$ for arm $a$ produces the policy input:

$$A_t = \pi_\theta(F_a(I_{\le t}, u_k), q_{\le t}), \qquad \Delta\theta = 0. \tag{1}$$

History is the native execution-segment history, reset on stage-instruction changes by the shared protocol. The policy is queried every 16 control steps and returns its original $16 \times 12$ action chunk; all native action coordinates and gripper commands are preserved. "Training-free" means no parameter updates for this interface; the checkpoints are pretrained and may be benchmark-adapted. Both arms share planning candidates, entity binding, visibility rules, verification logic, action interface, and the oracle mask computation, so only the presentation and a fixed legend differ. The point arm's rendering is byte-identical to the frozen point-cue (bounding-box/crosshair) controller of the completed study (Sec. 5.1).

**Identity and information access.** Entity handles are derived from simulator entities and parent relationships and persist within an episode through stage changes and occlusion: simulator-assisted identity retention, not learned re-identification. The geoms of the bound object or part, asset-authored placement anchors, and current poses, bounds, depth, calibration, and visibility construct the two-dimensional cues. This is an *oracle-assisted simulation track*, not RGB-only grounding or an observation-restricted leaderboard result; equal access in both arms controls their comparison but does not remove the extra information. Official success values and metric diagnostics are logged but never choose the online cue; an asset anchor can coincide with a scorer's reference even though its threshold is never read.

### 3.2. Single-sample geometry rendering

The mask augmenter owns a private MuJoCo segmentation/depth rendering context with zero offscreen samples (`offSamples=0`) in which the robosuite gripper visualization sites are hidden; it never writes to the policy RGB context. Two measured defects motivated this design (Sec. 5.2): with multisampled segmentation, decoded

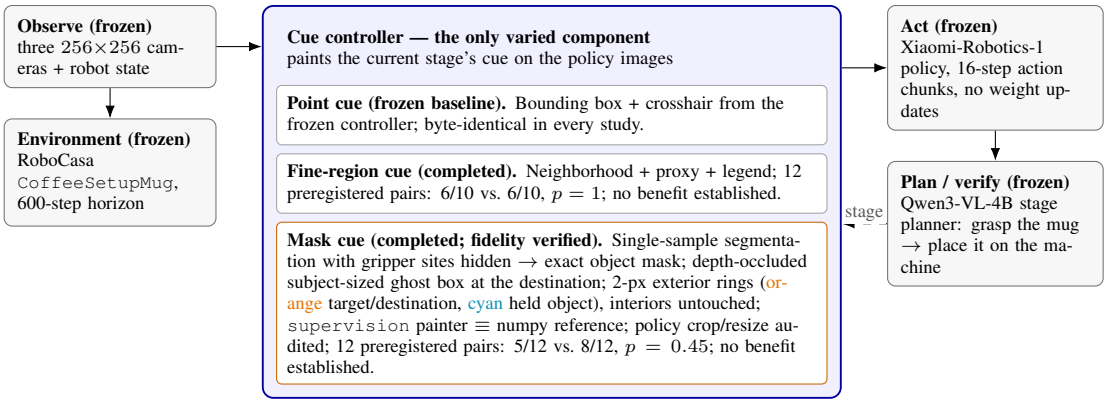



Figure 1. What varies and what stays frozen. Three 256×256 camera images and the robot state feed the frozen policy; the frozen Qwen3-VL-4B planner/verifier supplies the active stage. Only the cue controller differs between arms (point cue; fine-region cue, Sec. 5.1; MaskCue contour cue, Secs. 5.2 and 5.3); every arm shares the preregistered paired protocol of Sec. 4.

masks were inconsistent with the entity registry and the destination-proxy cross-check fell to IoU 0.763; and the gripper sites, present in segmentation but not in the policy RGB, hid up to 46% of the mug in the wrist camera. Depth for occlusion is taken from the same single-sample render, and the destination support height is anchored by a downward `mj_ray` query against the parent fixture. The evaluation cohort additionally zeroes the policy-RGB offscreen samples in *both* arms through a declared site override (Sec. 4) that touches only that XML attribute, not materials, physics, actions, or success thresholds.

For each camera and state, the augmenter obtains a sites-hidden segmentation $S_t^-$ and a with-sites reference $S_t^+$. An object's visible mask is the union of pixels whose segmentation type is geom and whose identifier belongs to that object (site identifiers never count as geom identifiers). For each object mask $M^-$ and reference $M^+$, drawability requires

$$M^+ \subseteq M^-, \qquad M^- \setminus M^+ \subseteq \{p : S_t^+(p) \text{ is a site}\}, \quad (2)$$

so hiding sites may reveal site-covered pixels but may not remove object pixels or add unrelated ones; failed checks suppress the contour. A registry bounding box is recorded as a diagnostic only.

### 3.3. Grasp and placement roles

During grasp, the primary target object or part receives an orange exterior contour; during placement, the held subject receives cyan and the bound destination orange. The held subject is resolved from the shared stage binding and prior grasp context, never from task success. For an object or part destination the cue follows its visible mask. For an abstract placement region, which has no pixels, we build a subject-sized geometric ghost: a box prism with the held subject's root-local bounding-box footprint and height, aligned with the parent fixture's axes, whose bottom rests on the support height under the region anchor. Camera rays are intersected analytically with this prism; a pixel belongs to the ghost mask if the ray's box-entry depth is visible within a fixed 3 mm occlusion tolerance against the single-sample depth buffer and the pixel is not part of the held subject's current mask. The proxy is a feasible-space visualization only, not the mug's exact shape, a semantic mouth-to-nozzle constraint, a collision-free certificate, or the scorer's acceptance region.

### 3.4. Exterior contours and protected interiors

Let $M_r$ denote the current mask of role $r$. Components with fewer than three pixels are excluded from contour generation but remain protected. Let $\widetilde{M}_r$ contain the retained eight-connected components, and let $P = \bigcup_r M_r$ include every role, drawable or not. A two-pixel exterior ring is generated from the Chebyshev dilation:

$$R_r = \left(\mathrm{dilate}_2(\widetilde{M}_r) \setminus \widetilde{M}_r\right) \setminus P. \quad (3)$$

Ring pixels inside any role mask are therefore never painted; when the held subject lies inside the destination proxy, its cyan ring is suppressed wherever it would cover proxy pixels. Overlaps between rings are resolved in a fixed order with the target or destination last. Colors are the frozen controller constants, orange $(255, 176, 0)$ for the target or destination and cyan $(0, 220, 255)$ for the held subject. Painting uses the supervision 0.27.0 `MaskAnnotator` at opacity 1.0 on a copy of the image (as an annotation library, not a segmentation model), and every painted frame is checked byte-for-byte against an independent NumPy painter. The unchanged-interior requirement covers all role masks, including those whose contours are suppressed; the point arm records how many interior pixels its box/crosshair changes but is not held to it.

**Fixed text intervention.** The grasp-stage legend is: "A thin orange outline drawn just outside this target traces its exact visible boundary; the outline never covers the target itself." The placement legend is: "A thin cyan outline traces the exact visible boundary of the held object, and a thin orange outline traces the visible space that the held object occupies when placed at the destination; neither outline covers the objects themselves." These are fixed instructions selected by stage operation, not measured precision claims; they establish neither exactness after the policy's preprocessing (Sec. 5.2) nor semantic validity of the proxy.

**Policy-input fidelity audit.** The frozen policy client center-crops each $256 \times 256$ image (crop ratio 0.95) and resizes it bilinearly, so raw-frame exteriority does not guarantee exteriority after interpolation. The audit executes that client function and measures, per camera and state, whether the ring survives the crop, whether any pixel of the *full-coverage* interior (bilinear mask coverage one) changed, and how many pixels inside the nearest-neighbor-resized mask changed, with their Chebyshev depth from the mask boundary. The subsequent model processor and final vision tensor are not inspected; a missing edge or mixed boundary is reported, not repaired by changing native preprocessing.

## 4. Evaluation Protocol

**Scope and comparison.** All studies use the RoboCasa task `CoffeeSetupMug` [5, 7], with three $256 \times 256$ RGB cameras (left and right agent views and the wrist camera), a 600-step native horizon, and the frozen policy and planner/verifier of Sec. 3. The point control is the existing hierarchical point-focus wrapper, not an unwrapped native policy; each treatment replaces its point/box cue with a different presentation plus a fixed explanatory sentence, so every comparison measures a combined cue-plus-legend intervention. The earlier experimental ALIGN/EXECUTE feedback variant is excluded from both studies.

**Paired design shared by both studies.** Twelve integer seeds, absent from all scanned local experiment records, are frozen in an immutable manifest before execution; each seed receives one episode per arm, the arm order alternates across pairs, and both arms of a pair run on the same host and renderer cohort. Before the second arm's first action, the complete initial-state signature (environment fingerprint, raw images, proprioception, entity catalog, and shared geometry augmentation) and an independently generated natural stage plan must equal the first arm's; a mismatch is an unknown invalid pair, the plan is not injected, and the seed is not replaced. Official success within the horizon is the primary outcome and reaching the horizon without success a known failure; infrastructure timeouts, missing or corrupt traces, invalid pairing, and unattempted cases remain unknown and are never converted to either. Completed episodes are not rerun.

**Registered analysis.** For $n$ admissible pairs with outcomes $s_i^T, s_i^P \in \{0, 1\}$ in the treatment and point arms, we report the paired risk difference $\widehat{\Delta} = \frac{1}{n}\sum_i (s_i^T - s_i^P) = (b - c)/n$, where $b$ counts treatment-only and $c$ point-only successes, the discordant counts, a two-sided exact McNemar test conditional on the discordant pairs (no discordant pair gives $p = 1$), a conservative paired risk-difference interval from Bonferroni-adjusted 97.5% Clopper–Pearson intervals for the win and loss probabilities, Wilson 95% intervals per arm, per-arm success/failure/unknown ledgers, and a worst-case envelope in which each unknown arm takes either outcome; the interval and the conditional test are not inversions of one another. With 12 pairs, even five wins and zero losses give exact two-sided $p = 0.0625$: the design can establish only large effects, and an inconclusive result does not prove equivalence. The secondary diagnostic, the terminal horizontal distance between the object root and the selected receiving site, is computed offline including failures and does not replace success.

**Completed fine-region study.** The treatment retained the point presentation and, during active placement only, added a $3 \times 3$ parent-frame lattice of single-pixel neighbor points around the region center, a radius-three outline at the held object's bounding-box center proxy, and a fixed legend. Development seeds 20261025–20261027 were excluded; seeds 20261031–20261042 were selected by a local history audit and frozen with the analysis plan before the first episode. All inference ran on one laptop RTX 5090 GPU under WSL with a single worker. A host shutdown interrupted the run after six episodes (complete pairs for seeds 20261031–20261033) while the seventh case (seed 20261034, treatment arm) had opened its trace with zero records; a resume driver written after registration executed the remaining 18 cases, including that case, with the same manifest, configuration, ports, and models. Under the registered rule an opened trace is unknown, so the primary analysis excludes that pair; the analysis including it is a disclosed deviation-dependent supplement. In the final case (seed 20261042, point arm) one planner/verifier request exceeded the 180-second client timeout after 544 of 600 steps; it is an unknown infrastructure timeout, was not repeated, and its pair is excluded.

**Rendering cohorts and repeatability.** Three host cohorts hold the evidence and are never pooled: the laptop RTX 5090 (historical rendering setting) for the completed fine-region study and the pre-fix mask diagnostic only; the

H800 for the fidelity run and the pilot launch lost with the instance; the shared H100 cluster for the fidelity re-run, the pilot, and the completed mask-versus-point study. On the H800, at a fixed development state with zero control steps and zero policy calls and unchanged positions, velocities, and simulator time, repeated renders of the simulator's default four-sample policy RGB differed by 2,019–4,833 pixels per camera comparison, so byte-equality of initial images, part of the pair-admission signature, could not hold; with offscreen RGB samples set to zero, three consecutive images were byte-identical in each camera. A zero-sample policy-RGB cohort for *both* arms was therefore declared before any registered episode; the admission threshold was not relaxed, the earlier failure was retained, and the fidelity run found the policy RGB unchanged by the mask context at every checked step. OpenGL debug error 1281 (an out-of-range material shininess value) appeared in every probe, including the zero-sample one; its cause and impact remain unexplained. Repeatability at one static state does not certify determinism across resets or motion, so pair admission still requires byte-equal initial images.

**Fidelity checks and pilot gates.** A simulator-in-the-loop fidelity check ("T3") runs on the evaluation host without model services: for each development seed, 100 zero-action and 40 scripted-motion steps plus four injected place-like mug positions (at, above, and beside the receiving site, and in the gripper) for test-only coverage of the placement cue. All twelve checks of Table 2 were declared before the run. The engineering pilot (three development seeds, six cases, orders point→masks, masks→point, point→masks) was gated by ten predeclared conditions G1–G10: frozen sources; box consistency; depth convention; GL IoU $\geq$ 0.98; zero interior pixels changed; complete traces; initial-state and plan match in every second arm; overhead $\leq$ 100 ms and memory; the analyzer reading all six traces; and author sign-off on contact sheets (Fig. 2). Pilot success counts were never a gate or evidence.

**Frozen mask-comparison protocol.** An unused-seed audit with the same rule as the completed study, additionally excluding the development seeds and seeds 20261031–20261042, selected seeds 20261044–20261055. The formal manifest (12 pairs, 24 cases, two workers on one H800, runtime admission `max_local_seconds` = 21,000; SHA-256 prefix `94d4721e1452`) was frozen on the instance on 2026-10-26 before the pilot, a departure from the written plan made so that dry runs could exercise the final manifest; no case of it was executed. After the instance was lost, the manifest was re-frozen locally from the same sources, seeds, and arguments and reproduced the recorded 12-character prefix of the registered digest (the full server digest was not recorded), so the frozen protocol is recoverable. The primary analysis is identical to the completed study with masks in place of neighborhoods, using a frozen analyzer (test suite in Sec. 5.2). A residency probe on the H800 (three policy/planner pairs, six resident models, no environment step) peaked at 59,497 MiB with at least 21,583 MiB free, supporting two workers.

**Execution on a shared H100 cluster.** After the H800 instance was lost (Sec. 5.3), and before any mask outcome existed, the study was deployed on a shared eight-GPU NVIDIA H100 PCIe node (driver 570.195.03) with the same hash-verified source snapshot, vendor archives, and seed assets; the Python environments were rebuilt from the same pinned lists (CPython 3.12.3, torch 2.8.0+cu128, FlashAttention 2.8.3) and the weights re-admitted file by file against the registered digests. Two GPUs with a pending driver recovery action returned corrupt depth buffers and were excluded; EGL and CUDA device indices were verified to coincide. To use the three healthy GPUs, the frozen drivers re-froze both manifests with three workers (one per GPU, beside another user's training job) before any case had completed: the pilot digest prefix changed from `df91a4d28570` to `7c89c95a12e7`, the formal prefix from `94d4721e1452` to `8266cc633102`, and the derived formal runtime admission from 21,000 to 14,600 s (eight cases per worker); seeds, pair indices, arm order, arms, horizon, and pair keys are identical case by case. Three launches produced no episode and are archived unchanged: a two-worker pilot stopped after three minutes to apply the re-freeze, and two launches the driver refused because a port of a stopped service was still occupied. The fidelity check was repeated on the cluster before the pilot (Sec. 5.2). A resume is admitted only when the prior launcher left no completion receipt, all prior results are task outcomes, the owed cases form complete pairs, at most one zero-step trace exists, and hashes of completed results and interrupted traces verify; none was needed on the cluster. Task time (first reset observation to terminal step, excluding reset, model startup, and post-terminal logging) is reported in Sec. 5.3.

## 5. Results

### 5.1. Point versus fine-region cues: a null result

A development stage with three episodes per arm (points 1/3, neighborhoods 2/3) motivated the validation and is not pooled into it. The validation was interrupted once and contains one re-executed zero-step case and one unknown infrastructure timeout (Sec. 4); Table 1 gives both the registered-rule primary analysis and the deviation-inclusive supplement.

Under the registered rule, 10 of the 12 planned pairs were admissible; both arms succeeded on the same six pairs

Table 1. Completed point-versus-neighborhood validation (12 planned seeds per arm). S/F/U are successes, known failures, and unknown cases; paired success and Wilson 95% intervals use the admitted seed pairs of each analysis. Not mask-policy results.

| Analysis | Arm | S/F/U | Paired | Wilson 95% (%) |
|---|---|---|---|---|
| Registered | Point | 7/4/1 | 6/10 | [31.3, 83.2] |
| | Neighborhood | 6/5/1 | 6/10 | [31.3, 83.2] |
| With rerun | Point | 7/4/1 | 7/11 | [35.4, 84.8] |
| | Neighborhood | 6/6/0 | 6/11 | [28.0, 78.7] |

and failed on the same four, so the neighborhood-minus-point difference is 0.0 percentage points, with conservative 95% interval $[-35.5, 35.5]$ and exact paired $p = 1$. Including the re-executed case (seed 20261034: point success, neighborhood failure) admits eleven pairs with one point-only win and zero neighborhood-only wins: difference $-9.1$ points, interval $[-45.7, 32.7]$, $p = 1$. Letting each unknown arm take either outcome gives worst-case envelopes of $[-16.7, 0.0]$ points under the registered rule and $[-16.7, -8.3]$ with the rerun; these are bounds, not confidence intervals. Neither analysis supports a success advantage for the fine-region presentation or establishes equivalence. The median within-pair terminal XY difference is $-0.31$ cm over the ten registered-rule pairs and $-0.30$ cm over the eleven (negative favors neighborhoods); this diagnostic includes failures and is not an accuracy guarantee.

### 5.2. Pixel fidelity of the mask interface on the H800 and the H100

The fidelity check ran on the evaluation instance on 2026-10-26 (22:00–22:09 UTC; NVIDIA H800 via EGL, driver 595.71.05; MuJoCo 3.3.1; supervision 0.27.0) with the zero-sample cohort active for both the policy and the mask context, on the three development seeds. Table 2 lists the twelve declared checks; eleven passed and the run's overall flag is *false*. The production masks agreed pixel for pixel with an independent reference render through the official robosuite render path (0 mismatched pixels for segmentation, object masks, depth, the analytic ghost, and the with-sites comparison; 189,225 object–camera comparisons). Registry diagnostics recorded 20,058 registry-visible entries (19,865 box-equal, 193 strict containments), 9 registry-hidden entries, 0 inconsistencies, 0 superset-violation pixels, and 0 non-site pixels gained by hiding sites.

**The failing check.** The destination proxy is computed analytically and cross-checked against an OpenGL render of the same box. Of 117 proxy masks, 115 were evaluated (two had no visible proxy pixel in the wrist camera for the in-gripper injected state), with minimum IoU 0.9925 and median 1.0, so gate G4 passed. The stricter declared check,

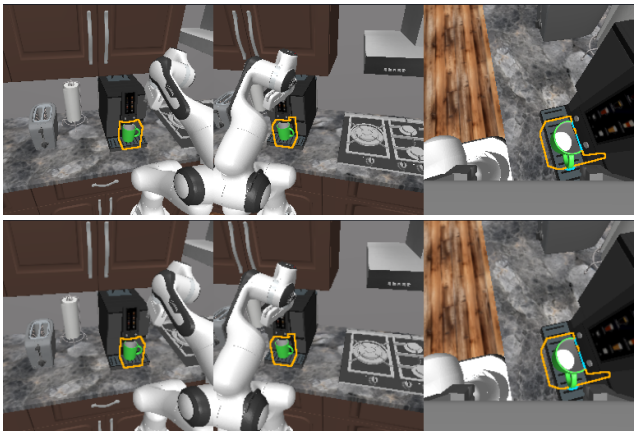


Figure 2. Unretouched rows from one contact sheet of the H800 fidelity run (seed 20261026; columns: left agent view, right agent view, wrist camera) at a place-like state with the mug *injected* at the receiving site for test coverage only, not a policy trajectory or task outcome. Orange outlines the visible part of the subject-sized destination proxy and cyan the held mug; in the agent views the mug lies inside the proxy, so its cyan ring is suppressed by Eq. (3). Top: the painted $256 \times 256$ policy image; bottom: after the frozen client's center crop and bilinear resize; both at $2\times$ nearest-neighbor magnification, rings two pixels wide in the source frame. Inspected by the authors (gate G10); no evidence about task success.

Table 2. Predeclared pixel-fidelity checks of MaskCue on the NVIDIA H800 (zero-sample cohort; three development seeds; 189,225 object–camera comparisons; 117 destination-proxy masks). Eleven of twelve checks passed. No code or threshold was changed after this run.

| Check | Result |
|---|---|
| Production mask context is single-sample | 0 samples; pass |
| Frozen extractor rule vs. registry (with sites) | 189,225/189,225; pass |
| Site hiding: superset rule, only site px gained | 0 / 0 violating px; pass |
| Pixel-identical to independent reference render | 0 mismatched px; pass |
| Depth convention equals frozen constant | legacy everywhere; pass |
| Ghost GL cross-check IoU $\geq 0.98$ (G4) | min 0.9925, med. 1.0; pass |
| Ghost GL cross-check: mismatches within 1-px band | 3 px outside, 1/117 masks; **fail** |
| Non-ghost pixels of ghost render unchanged | equal; pass |
| Inside-mask pixels changed by painting (G5) | 0; pass |
| Rings disjoint/exterior, ghost excludes subject, painter = reference | all true; pass |
| Policy RGB byte-identical with sites hidden | true; pass |
| Per-step overhead median $\leq 100$ ms (G8) | 74.6 / 72.7 / 70.2 ms; pass |

that every mismatched pixel lie within a one-pixel boundary band, failed: one scripted-motion mask (seed 20261026, step 128, wrist camera; analytic 2,767 px, GL 2,753 px, 14 mismatched px, IoU 0.9949) had 3 pixels outside the band; the other 116 had none. The policy sees the analytic ghost, which is the registered cue; the failure is recorded as a failure.

**Pre-fix diagnostic.** An earlier local run of the same check (2026-10-24/25, laptop RTX 5090 under WSL, D3D12), rendering masks from the four-sample policy context before the dedicated single-sample context existed, reached a minimum proxy IoU of 0.763 (median 0.934) with 7 pixels outside the band, 12,684 inconsistent entity–camera comparisons, 238,077 superset-violation pixels, 221,490 non-site pixels gained, and 276–301 ms median overhead; a no-multisampling variant in the same run reached minimum IoU 0.993 with zero violations. Its failing record is retained unchanged.

**Policy-input audit.** Over the 45 sampled camera states (three seeds $\times$ {initial grasp state, four injected place-like states} $\times$ three cameras, aggregated from the per-camera fidelity records), raw-frame fidelity was exact: 0 interior, 0 protected-mask, and 0 non-ring pixels changed. After the frozen client's center crop and bilinear resize, the ring remained visible in 45/45 states and no full-coverage interior pixel changed, but interpolation altered 21–388 pixels (median 89) inside the nearest-neighbor-resized mask in every state, all within the one-pixel inner boundary band, and in five wrist-camera states the crop removed 66–173 ring pixels (retention 0.73–0.96; 1.0 elsewhere). Exteriority thus survives at the policy input only as a boundary-band property; the legend's "never covers" wording is exact before, not after, the resize.

**Overhead, cluster re-run, and software checks.** Median per-step overhead of the full mask path (Table 2; augmenter 42.6/40.7/38.8 ms, place cue 32.0/32.0/31.4 ms) was within the 100 ms budget; both arms compute the masks, so this is the shared geometry cost, not the mask arm's increment over an unwrapped policy. The same check, sources, and cohort were run on a healthy GPU of the H100 node on 2026-10-27 before the pilot, with the same eleven of twelve outcomes: every count of Table 2 and of the registry diagnostics is reproduced exactly, including the same single mask with 3 pixels outside the band; median per-step overhead was 58.4, 46.0, and 45.9 ms. A first attempt on one of the two excluded GPUs had aborted at the depth-range check with an invalid depth value; that log is retained. The masks CPU test suite (synthetic inputs) passes 240 tests plus 39 subtests in WSL with supervision installed and 237 tests with three explicitly skipped supervision-dependent tests on Windows; it verifies software contracts, not rollouts.

### 5.3. Mask versus point cues: a second null result

**Lost first attempt and cluster pilot.** The engineering pilot was first launched on the rented H800 instance at about 23:07 UTC on 2026-10-26; about 20 minutes later the instance became unreachable (prepaid balance believed exhausted), no episode result, trace, or receipt was retrieved, and nothing from that attempt enters any count. On the H100 node the six-case pilot ran on 2026-10-27 with three workers. All six traces were complete and valid (G9), all 1,503 mask-presentation events reported zero interior pixels changed (G5), the 138 registered source files matched their digests (G1), and both arms of all three pairs passed the pre-action admission (G6–G7); G2–G4 and the overhead part of G8 come from the cluster fidelity run. The per-case memory part of G8 was not evaluated, and G10 (author sign-off) was recorded only for two H800 sheets and not re-recorded for the cluster, so the formal launch rested on the automated gates. Pilot success counts (point 2/3, mask 1/3) are not evidence.

**Registered 12-pair run.** The formal run executed on 2026-10-27 in 72 minutes (three workers, eight cases each, 217–630 s per case against a 1,600 s budget) with no infrastructure timeout and no resume. All 12 pairs are admissible: every second arm matched its first arm's initial signature and natural plan and no trace was missing, corrupt, or unknown, so the registered analysis and the worst-case envelope coincide. Table 3 gives the ledger: the point cue succeeded in 8 of 12 pairs and the mask cue in 5 of 12, a mask-minus-point risk difference of $-25.0$ percentage points with conservative 95% interval $[-74.1, 39.7]$ and exact paired $p = 0.453$ (seven discordant pairs). Every failure in either arm was a horizon exhaustion at 600 steps. The interval excludes neither a large harm nor a moderate benefit, so the result supports neither a benefit from exact contours nor equivalence.

**Secondary diagnostics.** Successes finished in similar time in both arms; the mask arm's longer median time to termination reflects its seven horizon exhaustions (Table 3). The terminal distance diagnostic, including failures, covers eleven cases per arm with a known value; the within-pair difference has median $+3.16$ cm (negative would favor masks). The outline was drawn on 5,281 of 5,630 mask-presentation events (86–99% of the steps of each stage and camera), so the policy did see the cue; the diagnostics describe the failures without explaining them. The frozen analyzer, run offline in the registered WSL runtime on the complete traces and again on the cluster, produced identical statistics; both reports, the ledger, receipts, re-frozen

Table 3. Completed mask-versus-point study on the H100 cluster (12 preregistered seed pairs, 24 cases, 0 unknown). Per-seed entries give the first and second arm in execution order with the outcome (S: success, F: horizon failure) and executed steps. Paired statistics are the registered contrast; terminal-XY and time rows are descriptive.

| Seed | First arm | Second arm | Seed | First arm | Second arm |
|---|---|---|---|---|---|
| 20261044 | point S 304 | mask F 600 | 20261050 | point S 312 | mask F 600 |
| 20261045 | mask F 600 | point F 600 | 20261051 | mask F 600 | point F 600 |
| 20261046 | point S 302 | mask S 317 | 20261052 | point S 257 | mask F 600 |
| 20261047 | mask F 600 | point S 308 | 20261053 | mask S 253 | point S 379 |
| 20261048 | point F 600 | mask S 323 | 20261054 | point S 263 | mask F 600 |
| 20261049 | mask S 222 | point S 215 | 20261055 | mask S 315 | point F 600 |

| Quantity | Point cue | Mask cue |
|---|---|---|
| Success / failure / unknown | 8 / 4 / 0 | 5 / 7 / 0 |
| Paired success; Wilson 95% (%) | 8/12; [39.1, 86.2] | 5/12; [19.3, 68.0] |
| Mask-only / point-only / both / neither | 2 / 5 / 3 / 2 | |
| Risk diff. (mask − point); conservative 95% | −25.0 points; [−74.1, 39.7] | |
| Exact paired McNemar $p$ | 0.453 | |
| Time to success, median (P90), s | 282 (304) | 277 (314) |
| Time to termination, all cases, median (P90), s | 300 (596) | 534 (615) |
| Terminal XY, median over known cases, cm | 2.14 ($n$=11) | 9.83 ($n$=11) |
| Pixel fidelity (Table 2; cluster re-run) | not re-measured | 11/12 pass |

manifests, probes, scripts, and a written disclosure of every deviation accompany the study.

## 6. Limitations

**Oracle information.** The masks, the held-subject identity, and the placement anchors come from the simulator, so the policy receives information beyond ordinary RGB observations; the study isolates presentation under shared access and establishes no learned segmentation, RGB-only grounding, real-world transfer, or compliance with an observation-restricted track.

**Geometry and image processing.** The destination ghost is a bounding-box-derived approximation rather than a semantic contact region, shaped by support estimation, root orientation, calibration, occlusion, and the 3 mm depth tolerance, and one of 117 proxy masks failed the one-pixel-band cross-check on both GPUs. An exterior ring bleeds into the one-pixel boundary band of the resized mask, the wrist-camera crop can remove part of a ring, and fidelity of the final model input tensor remains unverified.

**Causal scope.** Contours and the legend change jointly (as did the neighborhood points, proxy, and legend), the point control already includes hierarchical planning and verification, neither study compares against an unwrapped native policy, and no text-only, mask-only, box, or shuffled-cue ablation attributes an effect to geometric precision. The mask study shows that this presentation did not help, not why the mask arm exhausted the horizon more often (seven times versus four); the terminal-distance diagnostic is descriptive only.

**Rendering cohort and statistical scope.** The zero-sample cohort changes RGB antialiasing relative to the simulator default; applying it equally controls the within-cohort contrast but establishes neither unchanged policy performance nor benchmark comparability (Sec. 4). Each study concerns one task, one policy, one host, and 12 seeds; the conservative intervals span $\pm 35$ and $-74$ to $+40$ percentage points, so only large effects were detectable, and neither null result proves equivalence. Ordinary intervals do not correct for prompts changed after seeing outcomes, so each protocol was frozen first.

**Execution-site deviations.** The mask study's execution deviated from its registration (different host and GPU model, a lost first attempt, two excluded GPUs, shared GPUs, a three-worker re-freeze that changed digests and runtime admission but no seed, pair, order, or arm, the memory part of G8 not evaluated, G10 recorded for two H800 sheets only and not re-recorded on the cluster, two refused launches); Secs. 4 and 5.3 and the disclosure record list each.

## 7. Conclusion

Two preregistered paired studies on one RoboCasa task found no evidence that a more precise spatial cue changes the success of a frozen robot policy: fine-region cues tied the point cue (6/10 versus 6/10; $p = 1$), and exact object/destination contours succeeded less often (5/12 versus 8/12; $p = 0.45$) without reaching significance; neither study establishes equivalence (Sec. 5). The mask interface's pixel fidelity was verified on both evaluation GPUs with predeclared checks; the one failed check, the boundary bleed under the policy's resize, and every execution deviation of the mask study are reported as such (Secs. 4 and 5.3). The frozen protocols and this ledger of what was and was not measured are the contribution.

## LLM Usage Statement

LLM-based assistants (OpenAI Codex and Anthropic Claude Code) supported research planning and literature retrieval, software implementation, debugging and testing, analysis scripting and experiment operation, and manuscript drafting and editing under author direction. Numerical results were computed from recorded experimental and analysis outputs. The experimental models are described separately in Sec. 3. The authors are responsible for all claims; human visual inspection covered 2 of 12 contact sheets (Secs. 4 and 5.3).

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